Quick Start Guide
Picosecond Pulsed Fiber-Coupled Laser
PLD-XXX-SM-FC-U



1 About This Guide
This document provides a concise overview for installation and basic operation of the device. It does **not** replace the full product manual.

The complete manual is available at: https://flim-labs.github.io/docs/pld/PLD-manual.pdf

Scan the QR code to access the full manual.

Warning Read the full manual before operating the device. Improper use may result in hazardous laser exposure.

2 Safety Information
This product is classified as a Class 3R laser according to IEC EN CEI 60825-1 (ed. 3, 2014) and may emit radiation exceeding the maximum permissible eye exposure (MPE), posing a risk of serious eye injury when the beam is viewed directly.

Warning Visible and invisible laser radiation. Avoid direct eye exposure.

- Never look into the fiber output.
- Do not apply power without a connected and secured optical path.
- Do not point the laser at people or reflective surfaces.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.

3 Intended Use
The Picosecond Pulsed Fiber-Coupled Laser is intended for laboratory and research applications requiring pulsed or continuous laser excitation, such as fluorescence spectroscopy and FLIM measurements.

The device must be operated only by qualified personnel.

4 Product Overview
This laser is a compact fiber-coupled diode laser capable of continuous (CW) and pulsed operation. Different models are available each emitting at a different wavelength as listed in the table below.

Key Features
- Multiple wavelengths (405–850 nm)
- Fiber-coupled single-mode output
- CW and pulsed operation (5–80 MHz)
- External synchronization (SYNC IN)
- TTL synchronization outputs

Model	Wavelength
Visible radiation:	
PLD-405-SM-FC-U	405 nm
PLD-445-SM-FC-U	445 nm
PLD-488-SM-FC-U	488 nm
PLD-520-SM-FC-U	520 nm
PLD-635-SM-FC-U	635 nm
Invisible radiation:	
PLD-785-SM-FC-U	785 nm
PLD-808-SM-FC-U	808 nm
PLD-850-SM-FC-U	850 nm

5 Components Overview
Front panel



1. FC/PC output fiber connector
2. Laser emission ON/OFF button with status LED
3. Power selector (%): 20; 30; 40; 50; 55; 60; 65; 70; 75; 80; 90; 100
4. Operational mode selector: External Synchronization; Repetition Frequency; CW

Rear panel



1. Remote interlock
2. Power input (USB Type-C)
3. Status LED (POWER GOOD)
4. SYNC OUT 1 – synchronization output port 1
5. SYNC OUT 2 – synchronization output port 2
6. SYNC IN – synchronization input port

6 Scope of Delivery
- PLD laser module
- Optical fiber (single-mode)
- SMA cable (SYNC)
- USB Type-C power cable
- Documentation

7 Setup
Place the device on a stable, vibration-free laboratory surface with sufficient space for cable routing.

Before connecting the device, verify the following:
- The device and all cables are free of visible damage
- The optical fiber is clean and undamaged
- A suitable power source is available (5 V DC via USB Type-C, ≥ 3 A)

8 Connections
Optical Connection
- Connect the optical fiber to the FC/PC output
- Ensure that the fiber is properly seated and not under mechanical stress
- Ensure that the optical path is closed and secured against unintended exposure

Warning Laser emission may be enabled immediately when power is applied. Do not connect the device to the power source until the optical fiber is correctly connected and the optical path is secured.

Power Connection
- Connect the USB Type-C cable to the device
- Connect the power supply
- Verify the device status and laser emission status LEDs

Synchronization Interfaces
- SYNC IN: accepts an external TTL signal (2.5 V, 50 Ω) to trigger the laser
- SYNC OUT: provides a TTL signal synchronized with the laser emission

Notice When operating in *EXT SYNC IN* mode, the external trigger signal is routed to the SYNC OUT connector. The signal is transmitted whenever the laser is powered independent of laser emission status.

9 Operation
After completing all connections, the device is ready for operation. Select the desired operating mode and output power using the front panel controls, then press the laser emission ON/OFF button to enable emission.

Continuous Mode (CW)
- Set the operating mode selector to **CW**
- Adjust the output power using the power selector

Pulsed Mode (Internal Trigger)
- Select the desired repetition rate (5–80 MHz)
- Adjust the output power as required

External Trigger Mode
- Set the mode selector to **EXT SYNC IN**
- Apply an external TTL signal to the SYNC IN port
- The laser emission follows the external trigger signal

Shutdown
- Press the laser emission ON/OFF button to disable laser emission
- Verify that the emission status LED is off
- Disconnect the power supply if required

10 Further Information
Additional information and downloads are available on the product website: https://www.flimlabs.com/products/pulsed-laser-diode/

11 Product Label

Picosecond Pulsed Fiber-Coupled Laser

P/N: PLD-XXX-SM-FC-U
S/N: PLD-XXXX-XXX
5 V ⎓ 2.0 A | IP40

MADE IN ITALY - **FLIM LABS S.r.l.**
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12 Laser Classification Labels

VISIBLE LASER RADIATION
AVOID DIRECT EYE EXPOSURE
CLASS 3R LASER PRODUCT
Max average power ≤ 5 mW λ= 400-700 nm
STANDARD IEC EN 60825-1 Ed. 3 2014

INVISIBLE LASER RADIATION
AVOID DIRECT EYE EXPOSURE
CLASS 3R LASER PRODUCT
Max average power ≤2.9 mW λ= 785-850 nm
STANDARD IEC EN 60825-1 Ed. 3 2014

⚠ CAUTION
LASER
3R

13 Technical Specifications

Parameter	Value
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Maximum current consumption	2.0 A
Maximum power consumption	7.5 W
Optical and Laser Specifications	
Laser type	Solid-state diode laser
Available wavelengths in nm	405 | 445 | 488 | 520 | 635 | 785 | 808 | 850
Wavelength bandwidth	± 10 nm
Operating modes	CW or pulsed
Maximum average output power (pulsed))	1.2 to 2 mW wavelength dependent
Maximum output power (CW)	≤ 5 mW for 405 nm – 635 nm ≤ 2.9 mW for 785 nm – 850 nm
Power stability	< 2%
Pulse repetition frequency	5 MHz, 10 MHz, 20 MHz, 40 MHz, 80 MHz
Pulse duration (FWHM)	50 ps – 150 ps wavelength dependent
Output fiber core diameter	4 µm
M^2 value	1.06 *(for 4 µm core fiber)*
Polarization	Random
Optical output	FC/PC fiber connector
Signal Specifications	
Manual Controls	repetition frequency, output power
External synchronization input	TTL, 2.5 V, 50 Ω
Synchronization outputs	TTL, 2.5 V, 50 Ω
Physical Specifications	
Dimensions (L x W x H)	135 mm x 110 mm x 50 mm
Weight	550 g
Environmental Specifications	
Operating temperature	10°C to +40°C
Storage temperature	-20°C to +70°C
Relative humidity	80% non-condensing

Table 1: Key specifications of the Picosecond Pulsed Fiber-Coupled Laser.



Figure 1: Top view of the laser with indicated output.